

# Week 09 · When fine-tuning helps

OPEN / WEEK 09

The revised policy problem

Practice results say a tuned model wins 9/10, but a revised policy arrived yesterday and final cases show 6/10. What should change first?

Base



6/10

Prompt



8/10

Tuned



9/10

Practice scores only: test on fresh cases before choosing.

# First vote: what would you do?

## QUESTION / WEEK 09

Practice results say a tuned model wins 9/10, but a revised policy arrived yesterday and final cases show 6/10. What should change first?

Choose an initial answer and name one piece of evidence you need.

### VOTE FIRST • GIVE A REASON

**A Accept the first plausible result**

**B Check the source or measurement**

**Which source or test could settle this?**

# The journey through today's case

OPEN / WEEK 09

We will predict, connect an analogy, trace a mechanism, work an example, then explain a decision.

1

**Prompting**

2

**RAG**

3

**Fine-tuning**

4

**LoRA**

# 1. Predict · Prompting

## QUESTION / WEEK 09

When is this an inexpensive first trial?

Make a choice before we explain the diagram.

### VOTE FIRST · GIVE A REASON

**A When the failure concerns task instructions or format.**

**B A prompt can permanently teach yesterday's revised policy.**

**Which source or test could settle this?**

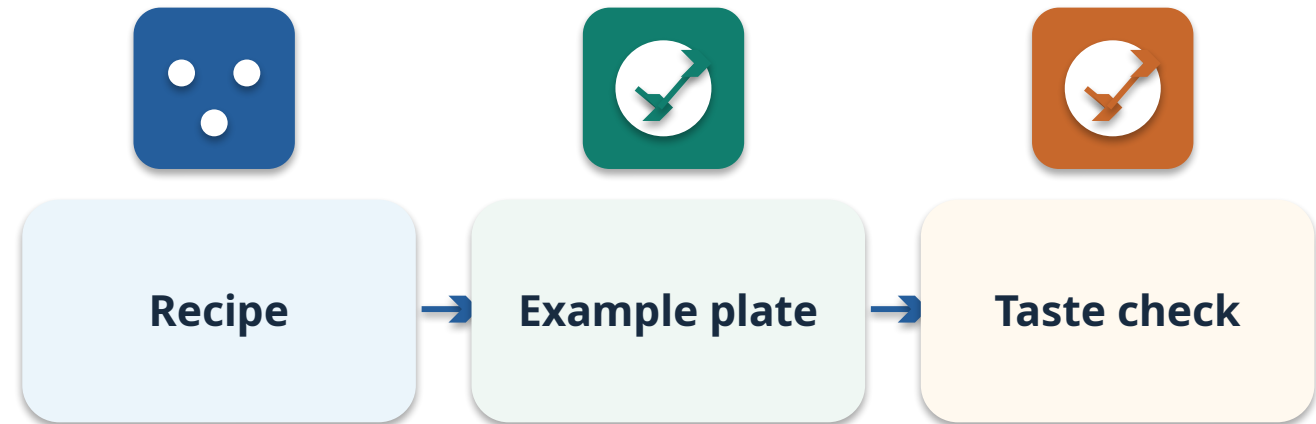
# Prompting · a familiar picture

## ANALOGY / WEEK 09

A clearer recipe can change a cook's output without retraining the cook.

What matches our case, and where does the analogy stop?

**A clearer recipe can change a cook's output without retraining the cook.**



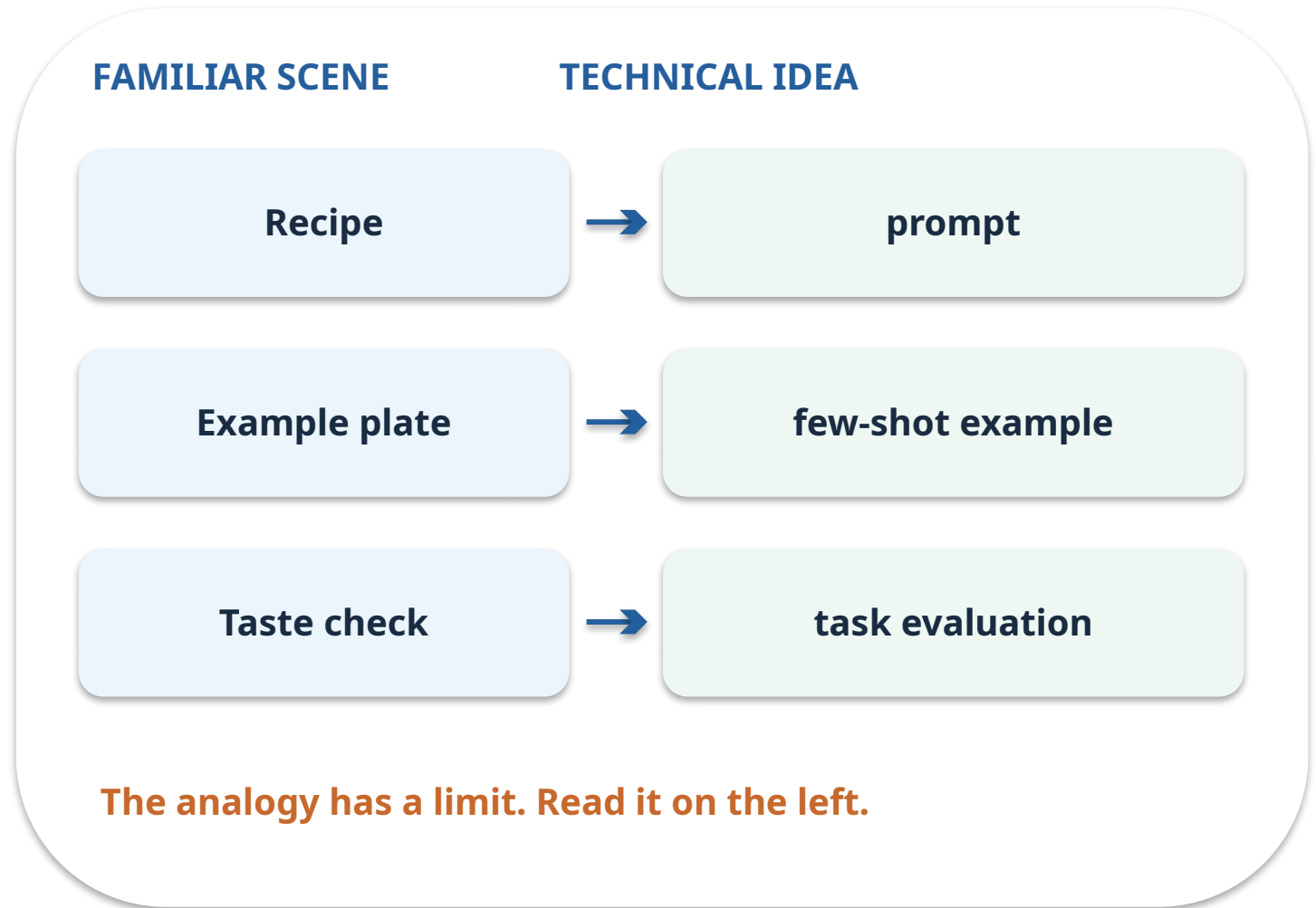
**What matches the technical process?**

# Prompting · map the parts

## MAPPING / WEEK 09

Match each everyday part to the technical part.

Analogy limit: Instructions can fail; a prompt does not update facts or guarantee conformance.



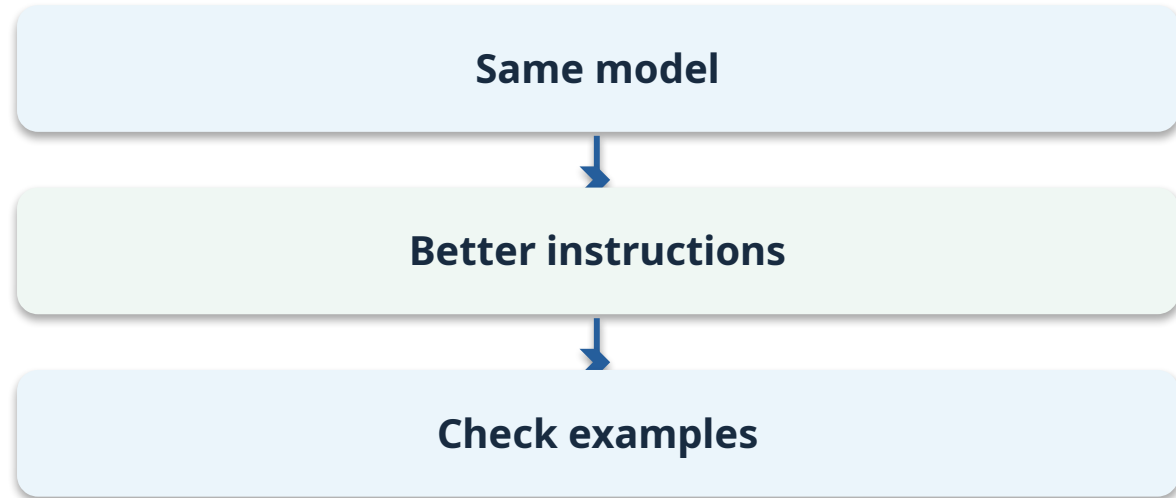
# Prompting · how it works

## MECHANISM / WEEK 09

Change instructions or examples for the task without changing weights.

Trace the arrows from left to right.

### FOLLOW THE DATA THROUGH EACH ARROW



# Prompting · work one case

## WORKED / WEEK 09

On ten practice cases, a clarified prompt improves 6/10 to 8/10; record the cases.

Point to the evidence before giving your conclusion.

### GIVEN

**On ten practice cases, a clarified prompt improves 6/10 to 8/10; record the cases.**

### QUESTION

**When is this an inexpensive first trial?**

### SUPPORTED ANSWER

**When the failure concerns task instructions or format.**

# Prompting · answer and explain

## REVEAL / WEEK 09

When the failure concerns task instructions or format.

Tempting mistake: A prompt can permanently teach yesterday's revised policy.

### TEST THE CLAIM AGAINST THE EVIDENCE

#### TEMPTING CLAIM

**A prompt can permanently teach yesterday's revised policy.**

#### SUPPORTED EXPLANATION

**When the failure concerns task instructions or format.**

Instructions can fail; a prompt does not update facts or guarantee conformance.

## 2. Predict · RAG

### QUESTION / WEEK 09

Where should a changed policy be updated first?

Make a choice before we explain the diagram.

### VOTE FIRST · GIVE A REASON

**A In the authoritative document and retrieval index.**

**B Retrieval changes model weights.**

**Which source or test could settle this?**

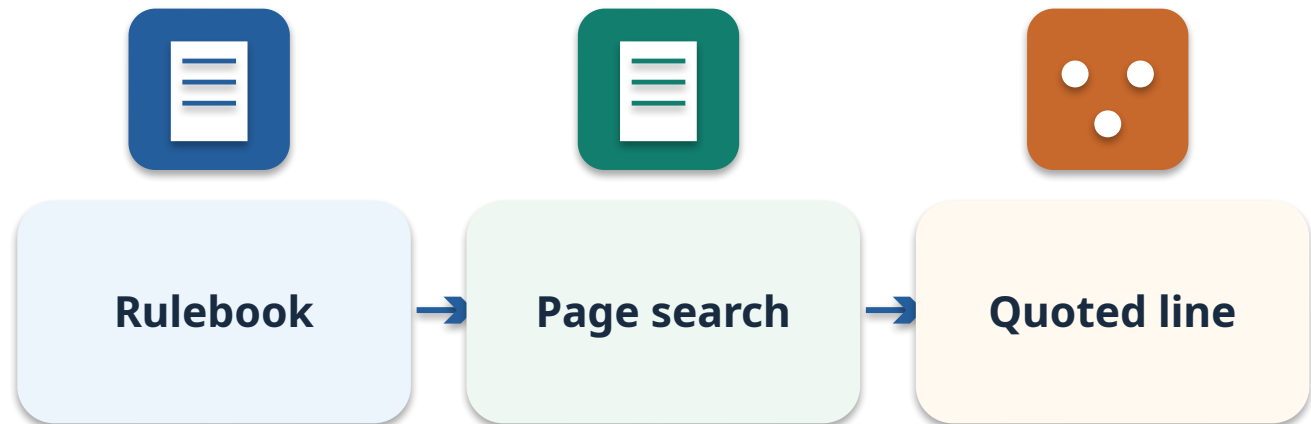
# RAG · a familiar picture

## ANALOGY / WEEK 09

A clerk opens the current rulebook instead of relying on memory.

What matches our case, and where does the analogy stop?

**A clerk opens the current rulebook instead of relying on memory.**



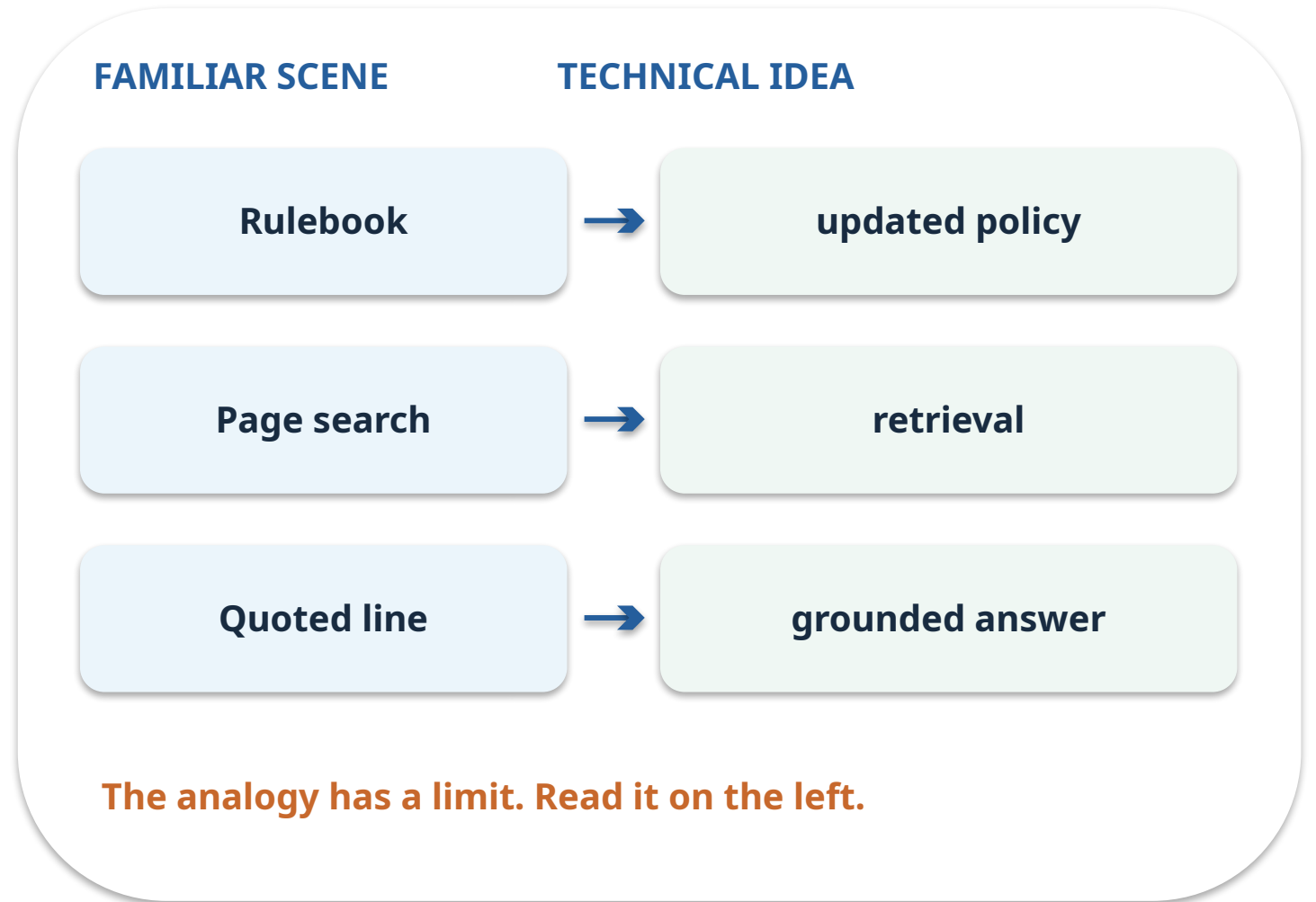
**What matches the technical process?**

# RAG · map the parts

## MAPPING / WEEK 09

Match each everyday part to the technical part.

Analogy limit: Retrieved text still needs access, relevance and claim checks.



# RAG · how it works

## MECHANISM / WEEK 09

Supply up-to-date evidence at inference time; useful for changing facts and traceable answers.

Trace the arrows from left to right.

### FOLLOW THE DATA THROUGH EACH ARROW



# RAG · work one case

## WORKED / WEEK 09

Replace P3 with its approved revised version, reindex and retrieve it for the question.

Point to the evidence before giving your conclusion.

### GIVEN

**Replace P3 with its approved revised version, reindex and retrieve it for the question.**

### QUESTION

**Where should a changed policy be updated first?**

### SUPPORTED ANSWER

**In the authoritative document and retrieval index.**

# RAG · answer and explain

## REVEAL / WEEK 09

In the authoritative document and retrieval index.

Tempting mistake: Retrieval changes model weights.

### TEST THE CLAIM AGAINST THE EVIDENCE

#### TEMPTING CLAIM

**Retrieval changes model weights.**

#### SUPPORTED EXPLANATION

**In the authoritative document and retrieval index.**

Retrieved text still needs access, relevance and claim checks.

# 3. Predict · Fine-tuning

## QUESTION / WEEK 09

Does the practice result prove tuning should ship?

Make a choice before we explain the diagram.

### VOTE FIRST · GIVE A REASON

**A No; the fresh test and failure analysis matter.**

**B Fine-tuning is guaranteed to beat a good prompt.**

**Which source or test could settle this?**

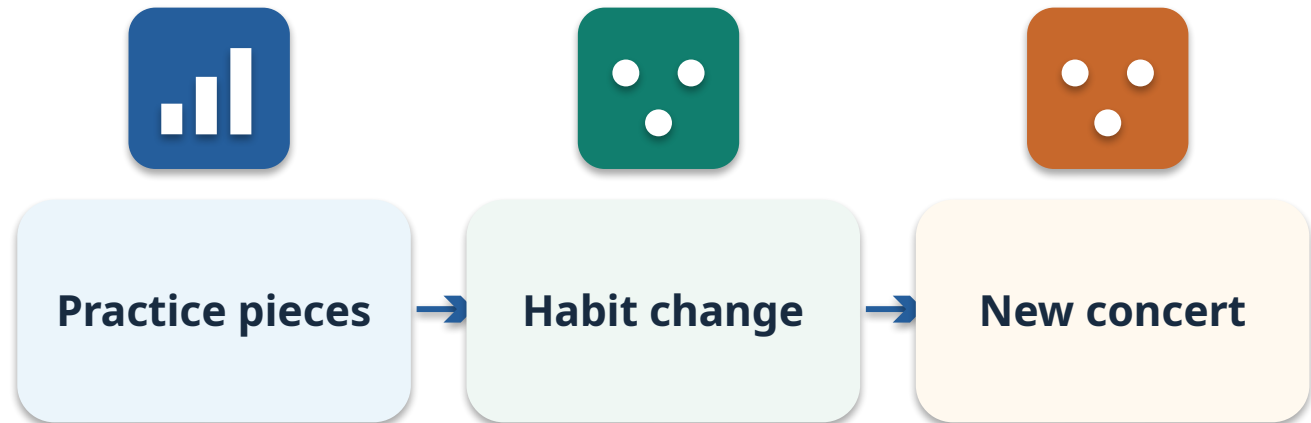
# Fine-tuning · a familiar picture

## ANALOGY / WEEK 09

A musician practises many examples to change a habit.

What matches our case, and where does the analogy stop?

**A musician practises many examples to change a habit.**



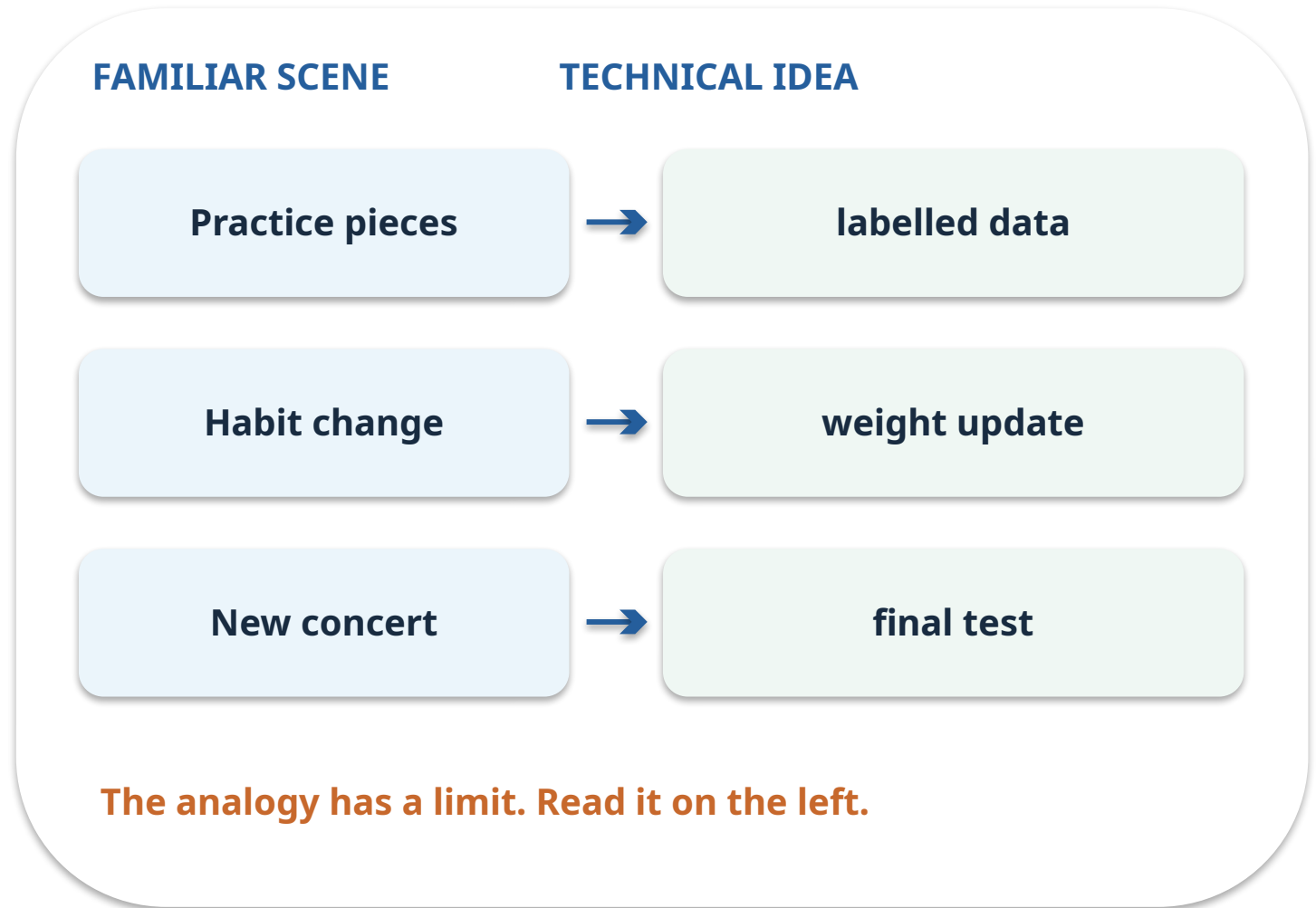
**What matches the technical process?**

# Fine-tuning · map the parts

## MAPPING / WEEK 09

Match each everyday part to the technical part.

Analogy limit: Small practice gains can reflect overfitting; tuning is not a live fact store.



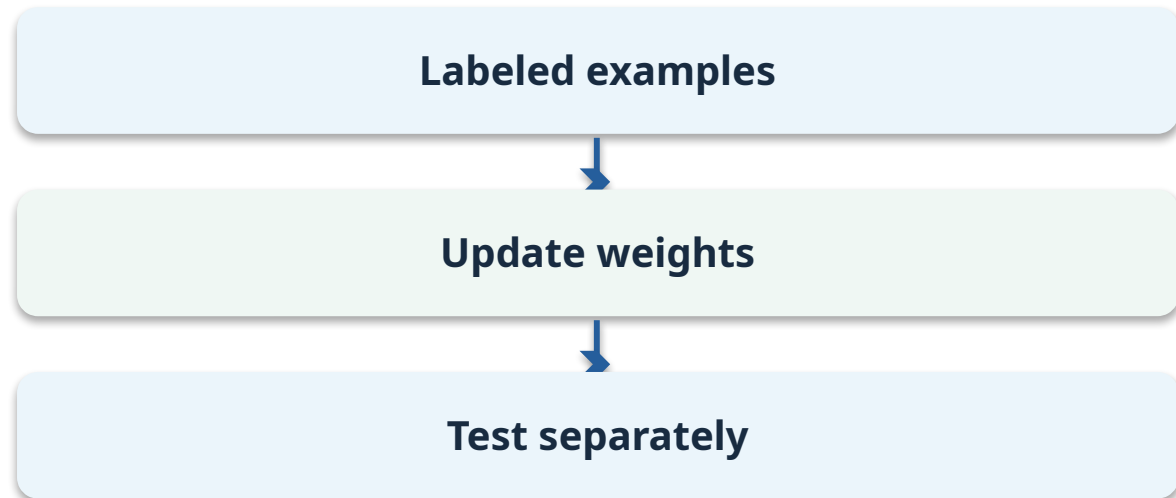
# Fine-tuning · how it works

## MECHANISM / WEEK 09

Train on examples to change learned behaviour; separate development from final test cases.

Trace the arrows from left to right.

### FOLLOW THE DATA THROUGH EACH ARROW



# Fine-tuning · work one case

## WORKED / WEEK 09

Base 6/10, prompted 8/10, tuned 9/10 in practice; final tuned 6/10. Investigate.

Point to the evidence before giving your conclusion.

### GIVEN

**Base 6/10, prompted 8/10, tuned 9/10 in practice; final tuned 6/10. Investigate.**

### QUESTION

**Does the practice result prove tuning should ship?**

### SUPPORTED ANSWER

**No; the fresh test and failure analysis matter.**

# Fine-tuning · answer and explain

REVEAL / WEEK 09

No; the fresh test and failure analysis matter.

Tempting mistake: Fine-tuning is guaranteed to beat a good prompt.

## TEST THE CLAIM AGAINST THE EVIDENCE

### TEMPTING CLAIM

**Fine-tuning is guaranteed to beat a good prompt.**

### SUPPORTED EXPLANATION

**No; the fresh test and failure analysis matter.**

Small practice gains can reflect overfitting; tuning is not a live fact store.

## 4. Predict · LoRA

### QUESTION / WEEK 09

What does LoRA reduce?

Make a choice before we explain the diagram.

#### VOTE FIRST · GIVE A REASON

**A The trainable parameter count compared with full fine-tuning.**

**B LoRA retrieves current policy documents automatically.**

**Which source or test could settle this?**

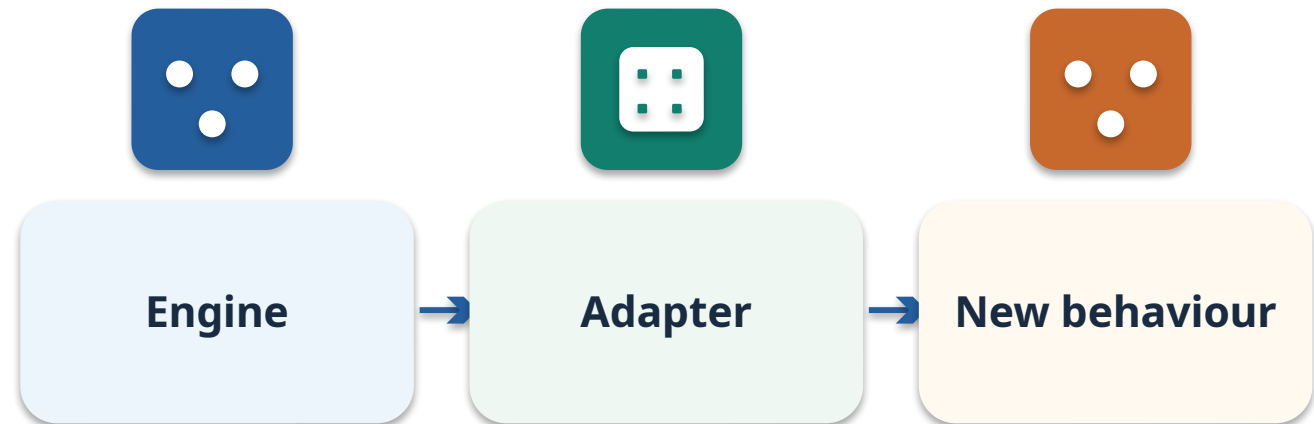
# LoRA · a familiar picture

## ANALOGY / WEEK 09

A small adapter changes a machine's behaviour without replacing its engine.

What matches our case, and where does the analogy stop?

**A small adapter changes a machine's behaviour without replacing its engine.**



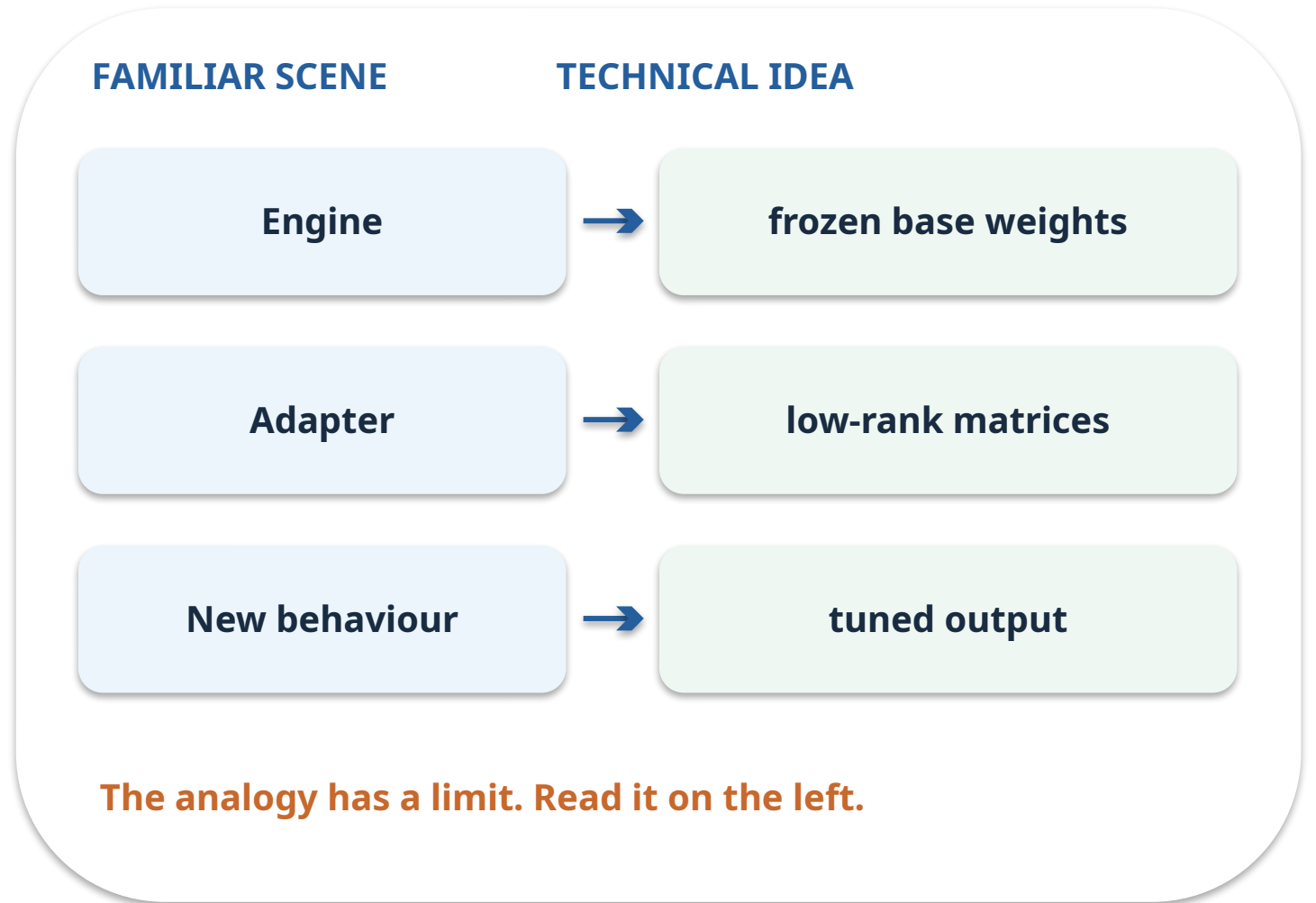
**What matches the technical process?**

# LoRA · map the parts

## MAPPING / WEEK 09

Match each everyday part to the technical part.

Analogy limit: Low trainable parameter count does not remove data, compute or evaluation needs.



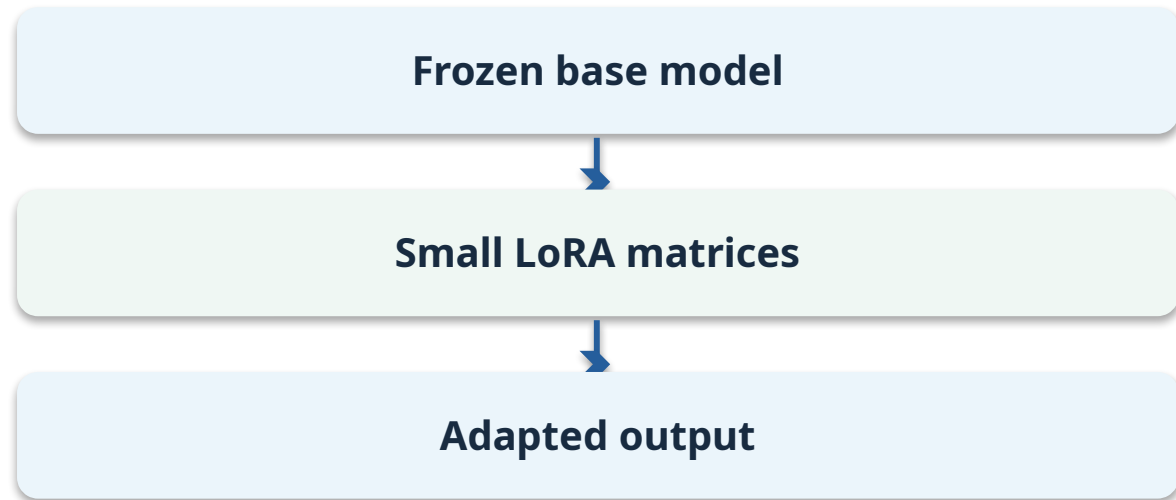
# LoRA · how it works

## MECHANISM / WEEK 09

Update a small trainable low-rank component instead of all model weights.

Trace the arrows from left to right.

### FOLLOW THE DATA THROUGH EACH ARROW



# LoRA · work one case

WORKED / WEEK 09

LoRA trains small matrices while freezing most base weights; no live tuning is required in this lab.

Point to the evidence before giving your conclusion.

**GIVEN**

**LoRA trains small matrices while freezing most base weights; no live tuning is required in this lab.**

**QUESTION**

**What does LoRA reduce?**

**SUPPORTED ANSWER**

**The trainable parameter count compared with full fine-tuning.**

# LoRA • answer and explain

REVEAL / WEEK 09

The trainable parameter count compared with full fine-tuning.

Tempting mistake: LoRA retrieves current policy documents automatically.

## TEST THE CLAIM AGAINST THE EVIDENCE

### TEMPTING CLAIM

**LoRA retrieves current policy documents automatically.**

### SUPPORTED EXPLANATION

**The trainable parameter count compared with full fine-tuning.**

Low trainable parameter count does not remove data, compute or evaluation needs.

# Evidence trail: input → result

## WORKED / WEEK 09

On fresh cases tuned score is 6/10.

The practice gain alone does not prove better generalization.

### GIVEN

**Base 6/10; prompt 8/10; tuned 9/10 on practice cases.**

### QUESTION

**On fresh cases tuned score is 6/10.**

### SUPPORTED ANSWER

**The practice gain alone does not prove better generalization.**

# Notebook: run the prepared example

## PRACTICE / WEEK 09

Analyze a supplied results pack without training a model.

Read the original output before editing.

1

**Compare provided base, prompted and adapted predictions**

2

**Score each on development cases**

3

**Inspect a held-out final set once**

4

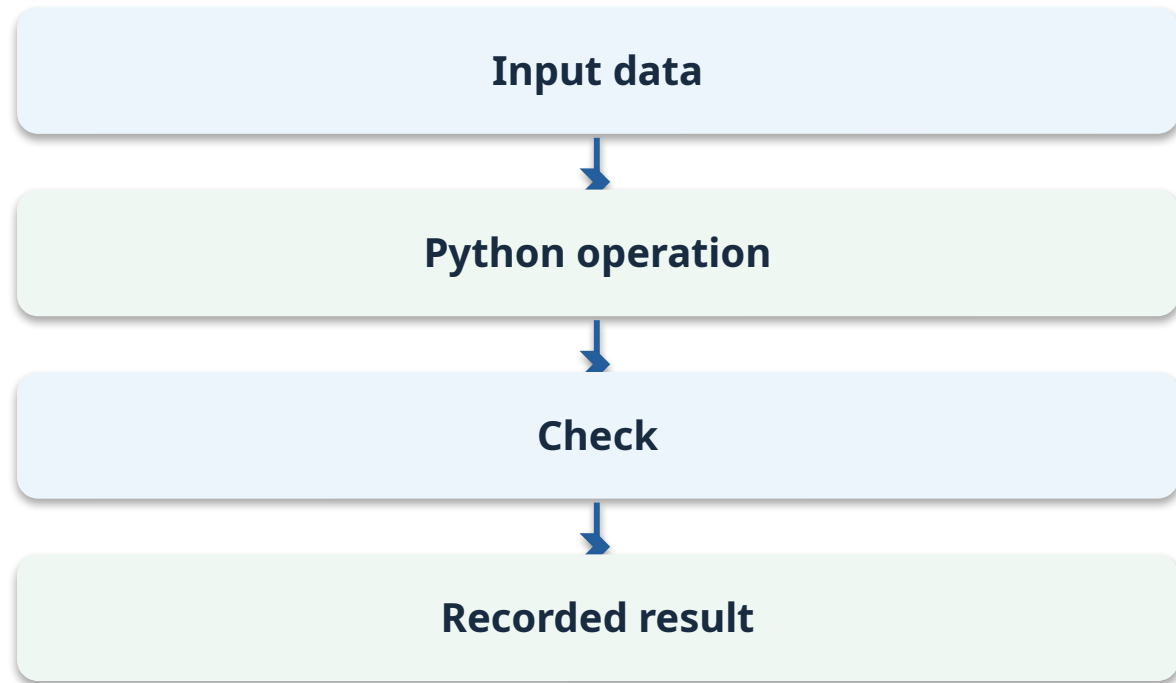
**Recommend a method for a policy update**

# Find the input and check in Python

## PRACTICE / WEEK 09

Locate the supplied input, the transformation, the checking line and the printed result.

### FOLLOW THE DATA THROUGH EACH ARROW



# Predict one controlled change

## PRACTICE / WEEK 09

Change one threshold and note why the final set must remain untouched.

What should change in the result? What stays fixed?

**Before: original case**

**After: one change**

**Keep all other inputs fixed.**

# Run, compare and record

PRACTICE / WEEK 09

Record the input, the one changed setting, the observed result and a reason for any difference.

|          |          |         |
|----------|----------|---------|
| Input    | Setting  | Result  |
| Original | baseline | record  |
| Changed  | one edit | compare |

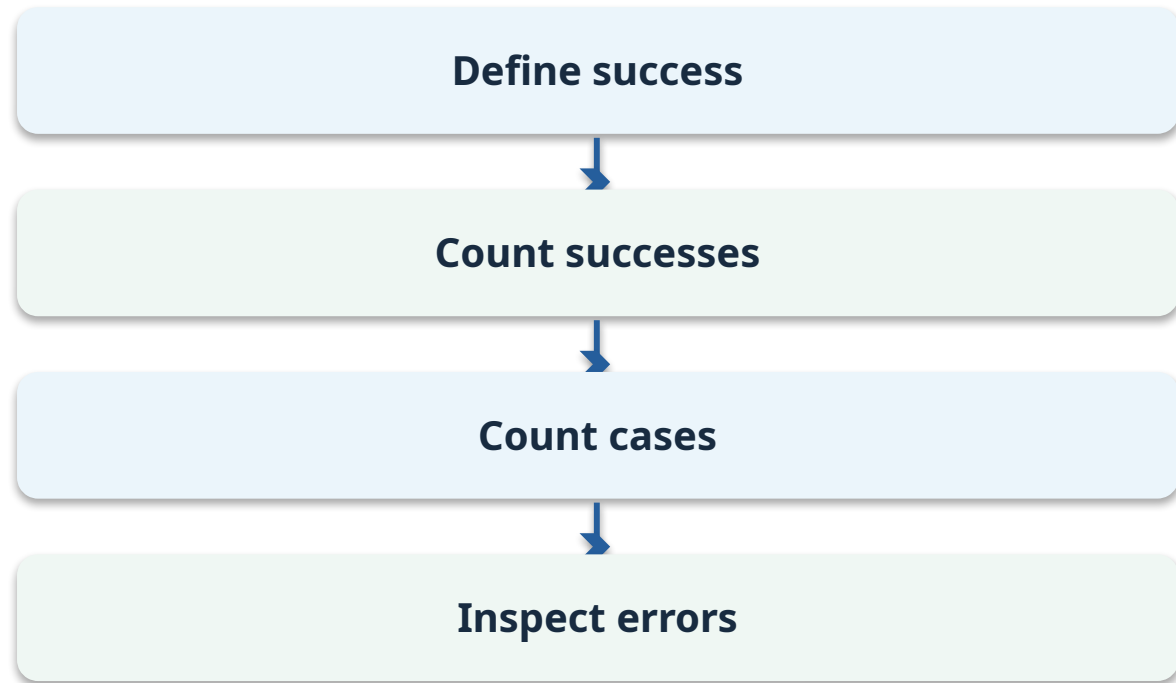
# Measure without hiding the denominator

## CHECK / WEEK 09

Accuracy with numerator and denominator on separate cases

Write the numerator, denominator and one failure case.

### FOLLOW THE DATA THROUGH EACH ARROW



# Error detective: find the layer

## CHECK / WEEK 09

Plausible wrong claim: Fine-tuning is the right way to add a policy changed yesterday.

Correct it using the case evidence.

### TEST THE CLAIM AGAINST THE EVIDENCE

#### TEMPTING CLAIM

**Fine-tuning is the right way to add a policy changed yesterday.**

#### SUPPORTED EXPLANATION

**Update the source document and retrieval index; weight adaptation is poor for rapidly changing facts.**

Name a test that would catch this failure.

# Transfer to a new situation

## CHECK / WEEK 09

The system returns the right fact but the wrong JSON field names. Which adaptation is the simplest first trial?

Explain which source or test would settle it.

### VOTE FIRST • GIVE A REASON

**A Use the visible evidence**

**B Flag missing or unsafe evidence**

**Which source or test could settle this?**

# Explain it to a partner

## CHECK / WEEK 09

Without reading the slide, name the input, one processing step and the check. Your partner should ask “How do you know?”

### FOLLOW THE DATA THROUGH EACH ARROW

**Base: 6/10**



**Prompt: 8/10**



**Fresh test: 6/10**

# Exam rehearsal · explain and apply

## CHECK / WEEK 09

Which method suits a policy that changed yesterday?

Write a short answer, then compare with the guide.

1

**State the core idea**

2

**Use a case detail or calculation**

3

**Add a limitation**

4

**Compare with the Word guide**

# Exam rehearsal · explain and apply

## CHECK / WEEK 09

What does LoRA reduce compared with full fine-tuning?

Write a short answer, then compare with the guide.

1

**State the core idea**

2

**Use a case detail or calculation**

3

**Add a limitation**

4

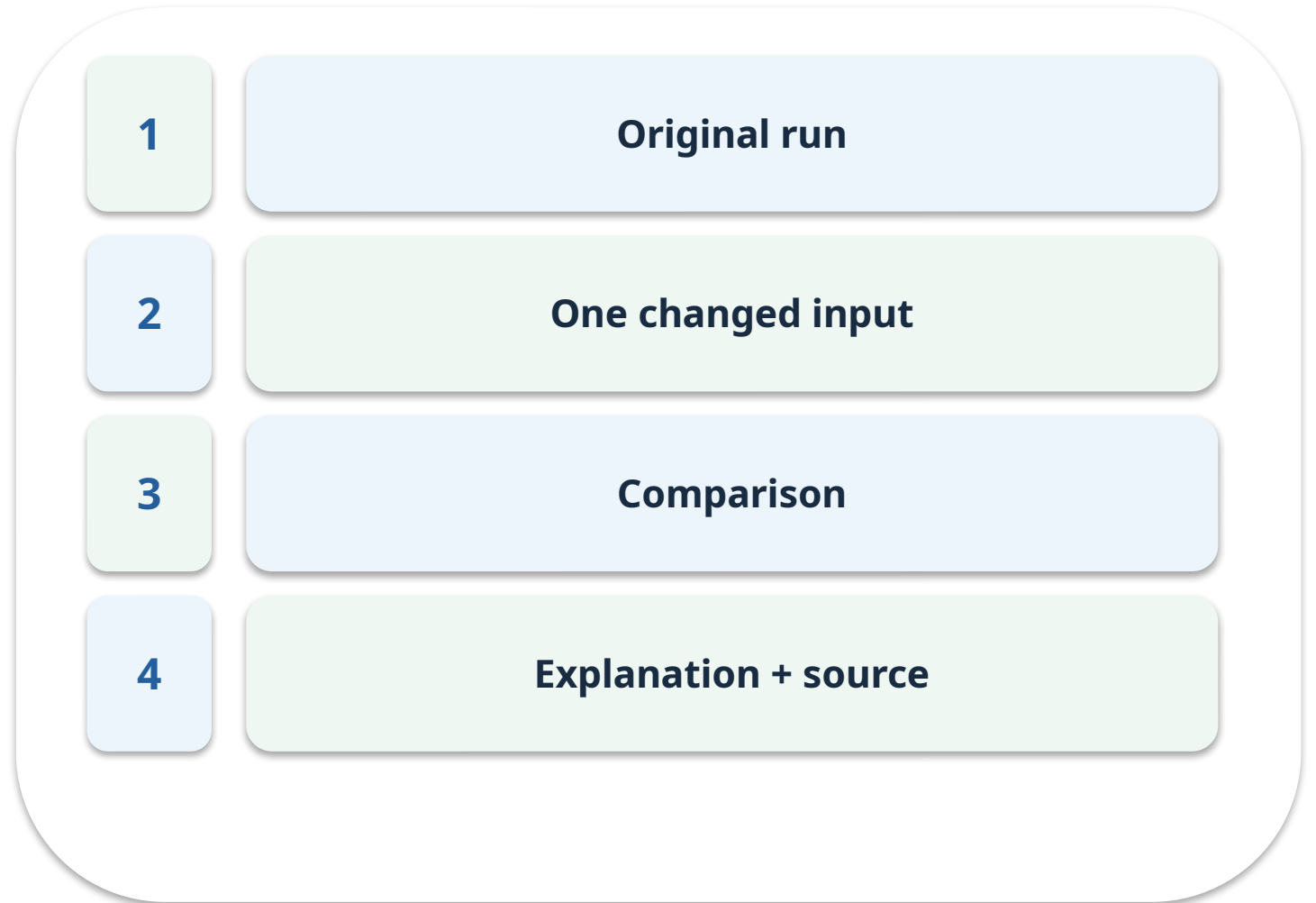
**Compare with the Word guide**

# What to hand in

## PRACTICE / WEEK 09

Comparison table with recommendation and uncertainty.

Include the one changed input and an error explanation.



# Three ideas to recall next week

## CHECK / WEEK 09

Diagnose the failure, choose prompt or RAG or tuning, and test fresh cases.

Read the linked official sources in the Word guide and redraw one diagram from memory.

1

**Prompting**

2

**RAG**

3

**Fine-tuning**

4

**LoRA**